25. Which one of the following is used as a standard for atomic mass.

(a) $_6C^{12}$ (b) $_7C^{12}$ (c) $_6C^{13}$ (d) $_6C^{14}$

II. Write brief answer to the following questions.

26) Define relative atomic mass.

27) What do you understand by the term mole.

28) Define equivalent mass.

29) What do you understand by the term oxidation number.

30) Distinguish between oxidation and reduction.

31) Calculate the molar mass of the following compounds.

i) Urea $[CO(NH_2)_2]$

ii) Acetone $[CH_3COCH_3]$

iii) Boric acid $[H_3BO_3]$

iv) Sulphuric acid $[H_2SO_4]$

32) The density of carbon dioxide is equal to 1.965 kgm^{-3} at 273 K and 1 atm pressure. Calculate the molar mass of CO_2.

33) Which contains the greatest number of moles of oxygen atoms

i) 1 mol of ethanol

ii) 1 mol of formic acid

iii) 1 mol of H_2O

34) Calculate the average atomic mass of naturally occurring magnesium using the following data

Isotope	Isotopic atomic mass	Abundance (%)
Mg^{24}	23.99	78.99
Mg^{25}	24.99	10.00
Mg^{26}	25.98	11.01

11th Std Chemistry 001-037.indd 33 23-12-2021 15:03:31